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Van der Corput inequalities for Bessel functions

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In this note, we offer some log-concavity properties of certain functions related to Bessel functions of the first kind and modified Bessel functions of the first and second kinds, by solving partially a recent conjecture on the log-convexity/log-concavity properties for modified Bessel functions of the first kind and their derivatives. Moreover, we give an application of the mentioned results by extending two inequalities of van der Corput to Bessel and modified Bessel functions of the first kind. Similar inequalities are proved also for modified Bessel functions of the second kind, as well as for log-concave probability density functions.

Keywords: Bessel functions of the first kind; modified Bessel functions of the first and second kinds; log-convexity; van der Corput inequality; probability density functions; trigonometricand hyperbolic functions

AMS Subject Classification: 33C10; 26A51; 26D07; 39B72

1. Introduction

For ν unrestricted real number, the modified Bessel function of the first kind of order ν, denoted usually by I_ν, is the particular solution of the differential equation [1, p.77]:

$$x^2y''(x) + xy'(x) - (x^2 + \nu^2)y(x) = 0. \tag{1.1}$$

The modified Bessel equation (1.1), which is of frequent occurrence in problems of mathematical physics, differs from the well-known Bessel's equation only in the coefficient of y. In fact, the modified Bessel function I_ν has the infinite series expansion [1, p.77]:

$$I_\nu(x) = \sum_{n\geq 0} \frac{(x/2)^{2n+\nu}}{n!\Gamma(\nu+n+1)} = \frac{(x/2)^\nu}{\Gamma(\nu+1)} \cdot {}_0F_1\left(\nu+1, \frac{x^2}{4}\right) \tag{1.2}$$

for all $\nu \neq -1, -2, \ldots$ and $x \in \mathbb{R}$, where Γ denotes the Euler gamma function and $x \mapsto {}_0F_1(\nu + 1, \frac{x^2}{4})$ is the hypergeometric function defined below. We note that the function $\mathcal{I}_\nu : \mathbb{R} \to [1, \infty)$,

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defined by

$$\mathcal{I}_\nu(x) = {}_0F_1\left(\nu+1, \frac{x^2}{4}\right) = 2^\nu \Gamma(\nu+1) x^{-\nu} I_\nu(x) = \sum_{n\geq 0} \frac{(1/4)^n}{(\nu+1)_n n!} \cdot x^{2n}, \tag{1.3}$$

for $\nu \neq -1, -2, \ldots$, (where for $a \neq 0$ the notation $(a)_n = a(a+1)(a+2)\cdots(a+n-1)$, $(a)_0 = 1$ is the well-known Pochhammer (or Appell) symbol) is strictly log-convex, when $\nu > -\frac{1}{2}$, see [2]. Observe also that, in particular, we obtain

$$\mathcal{I}_{-1/2}(x) = \sqrt{\frac{\pi}{2}} \cdot x^{1/2} I_{-1/2}(x) = \cosh x,$$

and thus actually $\mathcal{I}_\nu$ is log-convex for all $\nu \geq -\frac{1}{2}$. Moreover, recently it was conjectured that the function $x \mapsto \mathcal{I}_\nu(x)$ is strictly log-convex for all $\nu > -1$ and $x \in \mathbb{R}$, see [3, p.271]. We note that in [4], using Hölder's inequality for integrals, the authors claimed that all derivatives of the function $x \mapsto \mathcal{I}_\nu(x)$ are strictly log-convex for $\nu > -\frac{1}{2}$ and $x \in \mathbb{R}$. In [5] it was proved that in fact just the derivatives of even order of the function $\mathcal{I}_\nu$ will be strictly log-convex. Recall that $\mathcal{I}_{-1/2}(x) = \cosh x$, thus in this case it is clear that for all $k \in \{0, 1, \ldots\}$ the function $x \mapsto \mathcal{I}^{(2k)}_{-1/2}(x) = \cosh x$ is strictly log-convex on $\mathbb{R}$ and $x \mapsto \mathcal{I}^{(2k+1)}_{-1/2}(x) = \sinh x$ is strictly log-concave on $(0, \infty)$. This suggested the following conjecture (see [6, p.101], [7, p.187]).

The function $x \mapsto \mathcal{I}^{(2k)}_\nu(x)$ is strictly log-convex for all $k \in \{0, 1, \ldots\}$, $\nu > -1$ and $x \in \mathbb{R}$, while the function $x \mapsto \mathcal{I}^{(2k+1)}_\nu(x)$ is strictly log-concave for all $k \in \{0, 1, \ldots\}$, $\nu > -1$ and $x > 0$.

In this paper our aim is to give the full (negative) answer to the above conjectures for the case when $k = 0$. We will show that (see Section 4 (2)) there exists a $z_\nu \geq j_{\nu,1}$ (where $j_{\nu,1}$ is the first positive zero of the Bessel function J_ν) such that the function $x \mapsto \mathcal{I}_\nu(x)$ is strictly log-convex on $(0, z_\nu)$ and strictly log-concave on (z_ν, ∞). Moreover, we will prove that (see Theorem 1 and its proof) if $\nu \in (-1, -\frac{1}{2}]$, then the function $x \mapsto \mathcal{I}'_\nu(x)$ is strictly log-concave on $(0, \infty)$, while if $\nu > -\frac{1}{2}$, then there exists an x_ν, which satisfies $x_\nu > \sqrt{2(\nu+3)}$, such that $x \mapsto \mathcal{I}'_\nu(x)$ is strictly log-concave on $(0, x_\nu)$ and strictly log-convex on (x_ν, ∞).

The paper is organized as follows: in Section 2, we study the log-concavity of the function $x \mapsto \mathcal{I}'_\nu(x)$ and we apply the main result to extend an inequality of van der Corput concerning hyperbolic functions to modified Bessel functions of the first kind. Moreover, we give the counterparts of the above mentioned results for trigonometric functions and Bessel functions of the first kind. Motivated by these results, we present also the counterpart of these results for modified Bessel functions of the second kind. Section 3 contains the proofs of the theorems, and Section 4 is devoted for concluding remarks. In Section 4, among other things, we point out that the function $x \mapsto \mathcal{I}_\nu(x)$ cannot be strictly log-convex on $\mathbb{R}$ for all $\nu > -1$, and thus we disprove the conjecture from [3, p.271] (see also [6, p.100], [7, p.187]).

2. Inequalities for Bessel and modified Bessel functions

2.1. *Modified Bessel functions of the first kind*

Our first main result reads as follows.

THEOREM 2.1 *The following assertions are true:*

(a) If $\nu \in (-1, -\frac{1}{2}]$, then the function $x \mapsto \mathcal{I}'_\nu(x)$ is strictly log-concave on $(0, \infty)$.
(b) If $\nu > -1$, then the function $x \mapsto \mathcal{I}'_\nu(x)$ is strictly log-concave on $(0, \sqrt{2(\nu+3)})$.

(c) If $\nu > -\frac{1}{2}$, then there exists an x_ν, which satisfies $x_\nu > \sqrt{2(\nu+3)}$, such that $x \mapsto \mathcal{I}'_\nu(x)$ is strictly log-concave on $(0, x_\nu)$ and strictly log-convex on (x_ν, ∞).

The following inequality involving hyperbolic sine and cosine functions of van der Corput [8, p.270] are of special interest in this section:

$$|\cosh a - \cosh b| \geq |a-b|\sqrt{\sinh a \sinh b} \quad \text{for all } a, b \geq 0. \tag{2.1}$$

Taking into account the relation $\mathcal{I}_{-1/2}(x) = \cosh x$ the inequality (2.1) may be written as

$$|\mathcal{I}_{-1/2}(a) - \mathcal{I}_{-1/2}(b)| \geq |a-b|\sqrt{\mathcal{I}'_{-1/2}(a)\mathcal{I}'_{-1/2}(b)} \quad \text{for all } a, b \geq 0.$$

The following result is an extension of van der Corput's inequality (2.1).

COROLLARY 2.2 *If $a, b \geq 0$ and $\nu \in (-1, -\frac{1}{2}]$, then*

$$|\mathcal{I}_\nu(a) - \mathcal{I}_\nu(b)| \geq |a-b|\sqrt{\mathcal{I}'_\nu(a)\mathcal{I}'_\nu(b)}. \tag{2.2}$$

Moreover, the above inequality is valid for all $\nu > -1$ and $a, b \in (0, \sqrt{2(\nu+3)})$.

Since for all $n \in \{1, 2, \ldots\}$ and $\nu \neq -1, -2, \ldots$ one has $(\nu+1)(\nu+2)_{n-1} = (\nu+1)_n$, from (1.3) we have the following differentiation formula for the function $\mathcal{I}_\nu$, namely

$$2(\nu+1)\mathcal{I}'_\nu(x) = x\mathcal{I}_{\nu+1}(x), \tag{2.3}$$

where $x \in \mathbb{R}$ and $\nu \neq -1, -2, \ldots$. Now taking into account the relations $\mathcal{I}_{-1/2}(x) = \cosh x$ and

$$\mathcal{I}_{1/2}(x) = \sqrt{\frac{\pi}{2}} \cdot x^{-1/2} I_{1/2}(x) = \frac{\sinh x}{x},$$

the inequality (2.1) may be written as

$$|\mathcal{I}_{-1/2}(a) - \mathcal{I}_{-1/2}(b)| \geq |a-b|\sqrt{ab\mathcal{I}_{1/2}(a)\mathcal{I}_{1/2}(b)},$$

and from (2.3) it follows that the inequality (2.2) is equivalent to

$$2(\nu+1)|\mathcal{I}_\nu(a) - \mathcal{I}_\nu(b)| \geq |a-b|\sqrt{ab\mathcal{I}_{\nu+1}(a)\mathcal{I}_{\nu+1}(b)}. \tag{2.4}$$

We also note that, in particular, we have

$$\mathcal{I}_{3/2}(x) = 3\sqrt{\frac{\pi}{2}} \cdot x^{-3/2} I_{3/2}(x) = -3\left(\frac{\sinh x}{x^3} - \frac{\cosh x}{x^2}\right),$$

thus from (2.4) for $\nu = \frac{1}{2}$ we obtain the following interesting inequality

$$|b\sinh a - a\sinh b| \geq |a-b|\sqrt{(a\cosh a - \sinh a)(b\cosh b - \sinh b)} \quad \text{for all } a, b \in (0, \sqrt{7}).$$

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2.2. *Bessel functions of the first kind*

Now, let us consider the analogue of $\mathcal{I}_\nu$, that is, the function $\mathcal{J}_\nu : \mathbb{R} \to (-\infty, 1]$, defined by $\mathcal{J}_\nu(x) = 2^\nu \Gamma(\nu+1) x^{-\nu} J_\nu(x)$, where J_ν stands for the Bessel function of the first kind. Let us denote by $j_{\nu,n}$ the nth positive zero of the Bessel function of the first kind J_ν. The analogue of Theorem 2.1 is the following result.

THEOREM 2.3 *If $\nu > -1$, then the function $x \mapsto -\mathcal{J}'_\nu(x)$ is strictly log-concave on $(0, j_{\nu+1,1})$.*

The following inequality involving sine and cosine functions of van der Corput [8, p.237] is the analogue of (2.1):

$$|\cos a - \cos b| \geq |a-b|\sqrt{\sin a \sin b} \quad \text{for all } a, b \in [0, \pi]. \tag{2.5}$$

Taking into account the relation $\mathcal{J}_{-1/2}(x) = \cos x$ the inequality (2.5) may be written as

$$|\mathcal{J}_{-1/2}(a) - \mathcal{J}_{-1/2}(b)| \geq |a-b|\sqrt{\mathcal{J}'_{-1/2}(a)\mathcal{J}'_{-1/2}(b)} \quad \text{for all } a, b \in [0, \pi].$$

The following result is an extension of van der Corput's inequality (2.5).

COROLLARY 2.4 *If $a, b \in [0, j_{\nu+1,1}]$ and $\nu > -1$, then*

$$|\mathcal{J}_\nu(a) - \mathcal{J}_\nu(b)| \geq |a-b|\sqrt{\mathcal{J}'_\nu(a)\mathcal{J}'_\nu(b)}. \tag{2.6}$$

Observe that $\mathcal{J}_\nu$ satisfies the following differentiation formula

$$2(\nu+1)\mathcal{J}'_\nu(x) = -x\mathcal{J}_{\nu+1}(x), \tag{2.7}$$

where $x \in \mathbb{R}$ and $\nu \neq -1, -2, \ldots$. Now taking into account the relations

$$\mathcal{J}_{-1/2}(x) = \sqrt{\frac{\pi}{2}} \cdot x^{1/2} J_{-1/2}(x) = \cos x \quad \text{and} \quad \mathcal{J}_{1/2}(x) = \sqrt{\frac{\pi}{2}} \cdot x^{-1/2} J_{1/2}(x) = \frac{\sin x}{x},$$

the inequality (2.5) may be written as

$$|\mathcal{J}_{-1/2}(a) - \mathcal{J}_{-1/2}(b)| \geq |a-b|\sqrt{ab\mathcal{J}_{1/2}(a)\mathcal{J}_{1/2}(b)},$$

and from (2.7) it follows that the inequality (2.6) is equivalent to

$$2(\nu+1)|\mathcal{J}_\nu(a) - \mathcal{J}_\nu(b)| \geq |a-b|\sqrt{ab\mathcal{J}_{\nu+1}(a)\mathcal{J}_{\nu+1}(b)}. \tag{2.8}$$

We also note that, in particular,we have

$$\mathcal{J}_{3/2}(x) = 3\sqrt{\frac{\pi}{2}} \cdot x^{-3/2} J_{3/2}(x) = 3\left(\frac{\sin x}{x^3} - \frac{\cos x}{x^2}\right),$$

thus from (2.8) for $\nu = \frac{1}{2}$, we obtain the following inequality

$$|b\sin a - a \sin b| \geq |a-b|\sqrt{(a\cos a - \sin a)(b\cos b - \sin b)} \quad \text{for all } a, b \in [0, j_{3/2,1}],$$

where $j_{3/2,1} = 4.493409457\cdots$ is the first positive root of the equation $\sin x = x \cos x$.

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2.3. *Modified Bessel functions of the second kind*

For $\nu > 0$ let us consider the function $\mathcal{K}_\nu : (0,\infty) \to (0,\infty)$, defined by $\mathcal{K}_\nu(x) = 2^{\nu-1}\Gamma(\nu)x^\nu K_\nu(x)$, where K_ν is the modified Bessel function of the second kind, called sometimes as the MacDonald function. The analogue of Theorem 2.1 is the following result.

THEOREM 2.5 *If $\nu \geq \frac{3}{2}$, then the function $x \mapsto -\mathcal{K}'_\nu(x)$ is strictly log-concave on $(0,\infty)$.*

The following result is an analogue of van der Corput's inequality (2.1) for modified Bessel functions of the second kind.

COROLLARY 2.6 *If $a, b \geq 0$ and $\nu \geq \frac{3}{2}$, then*

$$|\mathcal{K}_\nu(a) - \mathcal{K}_\nu(b)| \geq |a-b|\sqrt{\mathcal{K}'_\nu(a)\mathcal{K}'_\nu(b)}. \tag{2.9}$$

We note that, in particular,we have

$$\mathcal{K}_{3/2}(x) = \sqrt{\frac{\pi}{2}} \cdot x^{3/2}K_{3/2}(x) = \frac{\pi}{2}(x+1)\mathrm{e}^{-x}, \tag{2.10}$$

thus from (2.9) for $\nu = \frac{3}{2}$, we obtain the following inequality:

$$|(a+1)\mathrm{e}^{-a} - (b+1)\mathrm{e}^{-b}| \geq |a-b|\sqrt{ab}\mathrm{e}^{-(a+b)/2} \quad \text{for all } a, b \geq 0.$$

3. Proofs of the main results

Proof of Theorem 2.1. (a) and (b) First observe that for $\nu > -1$, the power series

$$\mathcal{I}'_\nu(x) = \sum_{n\geq 1} \frac{(1/4)^{n-1}}{2(\nu+1)_n(n-1)!} \cdot x^{2n-1}$$

has only positive coefficients, and thus it makes sense to study the log-concavity of $x \mapsto \mathcal{I}'_\nu(x)$. Moreover, observe that $x \mapsto \mathcal{I}'_\nu(x)$ is strictly log-concave on $(0,\infty)$ if and only if $x \mapsto \mathcal{I}''_\nu(x)/\mathcal{I}'_\nu(x)$ is strictly decreasing on $(0,\infty)$. By using the recurrence formula [1, p.79] $[x^{-\nu}I_\nu(x)]' = x^{-\nu}I_{\nu+1}(x)$, we have

$$\mathcal{I}'_\nu(x) = 2^\nu\Gamma(\nu+1)x^{-\nu}I_{\nu+1}(x). \tag{3.1}$$

On the other hand application of [1, p.79] $xI_{\nu+1}(x) = xI'_\nu(x) - \nu I_\nu(x)$ and (3.1) yields

$$\mathcal{I}''_\nu(x) = 2^\nu\Gamma(\nu+1)x^{-\nu}\left[I_{\nu+2}(x) + \frac{1}{x}I_{\nu+1}(x)\right]. \tag{3.2}$$

By using the notation $R_\nu(x) = I_{\nu+1}(x)/I_\nu(x)$, in view of (3.1) and (3.2), we obtain that $x \mapsto \mathcal{I}''_\nu(x)/\mathcal{I}'_\nu(x)$ is strictly decreasing on $(0,\infty)$ for all $\nu \in (-1,-\frac{1}{2}]$ if and only if the function $x \mapsto Q_{\nu+1}(x)$, where $Q_\nu(x) = 1/x + R_\nu(x)$, is strictly decreasing on $(0,\infty)$ for all $\nu \in (-1,-\frac{1}{2}]$. To prove this we need to show that Q_ν is strictly decreasing on $(0,\infty)$ for all $\nu \in (0,\frac{1}{2}]$. However, since the function

$$x \mapsto Q_{1/2}(x) = \frac{1}{x} + R_{1/2}(x) = \frac{1}{x} + \frac{I_{3/2}(x)}{I_{1/2}(x)} = \coth x$$

is strictly decreasing on $(0,\infty)$, we can suppose that $\nu \in (0,\frac{1}{2})$. By using the fact (see [9, p.446]) that $R_\nu(x) \to 0$ as $x \to 0$, we have that $Q_\nu(x) \to \infty$ as $x \to 0$. On the other hand, by using the

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asymptotic expansion of $I_\nu(x)$ for large values of x, it can be seen (see [9, p.446]) that $R_\nu(x) \to 1$ as $x \to \infty$, and consequently $Q_\nu(x) \to 1$ as $x \to \infty$. Thus, for small values of x, the function Q_ν is strictly decreasing. Moreover, we can show that Q_ν is strictly decreasing on $(0, \sqrt{2\nu+4})$, and consequently on $(0, 2)$, for all $\nu > 0$. To prove this, first observe that $Q_\nu(x)$ can be rewritten as

$$Q_\nu(x) = \frac{I'_\nu(x)}{I_\nu(x)} + \frac{1-\nu}{x}$$

and by using twice the inequality [10, p.526]

$$y_\nu(x) = \frac{xI'_\nu(x)}{I_\nu(x)} > \nu + \frac{x^2}{\nu + 1/2 + \sqrt{x^2 + (\nu+3/2)^2}},$$

where $\nu > 0$, $x > 0$, we obtain that

$$\begin{aligned}
x^2 Q'_\nu(x) &= xy'_\nu(x) - y_\nu(x) + \nu - 1 \\
&= x^2 + \nu^2 - y_\nu^2(x) - y_\nu(x) + \nu - 1 \\
&< -\left(\frac{2\nu+1}{\nu + 1/2 + \sqrt{x^2 + (\nu+3/2)^2}} - 1 \right) x^2 - \frac{x^4}{(\nu + 1/2 + \sqrt{x^2 + (\nu+3/2)^2})^2} - 1 \\
&= \frac{2(\nu+1)x^2}{(\nu + 1/2 + \sqrt{x^2 + (\nu+3/2)^2})^2} - 1 \\
&= \frac{\sqrt{x^2 + (\nu+3/2)^2} - (\nu+1/2)}{\sqrt{x^2 + (\nu+3/2)^2} + (\nu+1/2)} \cdot \frac{2(\nu+1)x^2}{2(\nu+1) + x^2} - 1 \equiv w_\nu(x).
\end{aligned}$$

Now, taking into account that for $\nu > -\frac{1}{2}$ both functions

$$t \mapsto \frac{\sqrt{t + (\nu+3/2)^2} - (\nu+1/2)}{\sqrt{t + (\nu+3/2)^2} + (\nu+1/2)} \quad \text{and} \quad t \mapsto \frac{2(\nu+1)t}{2(\nu+1) + t}$$

are increasing on $(0, \infty)$, it follows that their product is also increasing and this in turn implies that for $\nu > 0$ and $x \in (0, \sqrt{2\nu+4})$ we have

$$w_\nu(x) < \frac{\sqrt{2\nu+4 + (\nu+3/2)^2} - (\nu+1/2)}{\sqrt{2\nu+4 + (\nu+3/2)^2} + (\nu+1/2)} \cdot \frac{2(\nu+1)(2\nu+4)}{2(\nu+1) + 2\nu + 4} - 1 = \varphi(\nu) < \lim_{\nu \to \infty} \varphi(\nu) = 0.$$

Here we used that the function φ is increasing on $(0, \infty)$. Summarizing, we have proved that for $\nu > 0$ and $x \in (0, \sqrt{2\nu+4})$, the function Q_ν satisfies $x^2 Q'_\nu(x) < 0$, which implies that Q_ν is strictly decreasing on $(0, 2)$ for $\nu > 0$. Thus, the first extreme of the function Q_ν, if any, is a minimum. Now, since $R_\nu(x)$ satisfies the differential equation:

$$u'(x) = 1 - u^2(x) - \frac{2\nu+1}{x} u(x),$$

it follows that $Q_\nu(x)$ satisfies the differential equation:

$$v'(x) = 1 - v^2(x) - \frac{2\nu-1}{x} v(x) + \frac{2\nu-1}{x^2}.$$

Differentiating both sides of the above relation, we obtain

$$v''(x) = \frac{2\nu-1}{x^2} \left[v(x) - \frac{2}{x} \right] - \left[2v(x) + \frac{2\nu-1}{x} \right] v'(x).$$

This shows that if $Q'_\nu(x_\star)=0$ for some $x_\star\geq 2$, then for that $x_\star$ we have

$$Q''_\nu(x_\star)=\frac{2\nu-1}{x_\star^2}\left[Q_\nu(x_\star)-\frac{2}{x_\star}\right]=\frac{2\nu-1}{x_\star^2}\left[R_\nu(x_\star)-\frac{1}{x_\star}\right].$$

On the other hand, by using the inequality [11, p.241]

$$R_\nu(x)>\frac{x}{\sqrt{x^2+(\nu+1)^2}+\nu+1},$$

which holds for $\nu\geq 0$ and $x>0$, we obtain that for $\nu\in(0,\frac{1}{2})$ and $x\geq 2$ we have

$$R_\nu(x)>\frac{x}{\sqrt{x^2+9/4}+3/2}\geq\frac{1}{x}.$$

This implies that $Q''_\nu(x_\star)<0$. But, this is in contradiction with the fact that the first extreme is a minimum, and consequently the derivative of Q_ν does not vanish when $\nu\in(0,\frac{1}{2})$. This in turn implies that Q_ν is strictly decreasing on $(0,\infty)$ for $\nu\in(0,\frac{1}{2})$, that is, $Q_{\nu+1}$ is strictly decreasing on $(0,\infty)$ for $\nu\in(-1,-\frac{1}{2})$. This is equivalent to the fact that if $\nu\in(-1,-\frac{1}{2})$, then the function $x\mapsto\mathcal{I}'_\nu(x)$ is strictly log-concave on $(0,\infty)$. On the other hand, since Q_ν is strictly decreasing on $(0,\sqrt{2(\nu+2)})$ for $\nu>0$, we obtain that $Q_{\nu+1}$ is strictly decreasing on $(0,\sqrt{2(\nu+3)})$ for $\nu>-1$, and this is equivalent to the fact that if $\nu>-1$, then the function $x\mapsto\mathcal{I}'_\nu(x)$ is strictly log-concave on $(0,\sqrt{2(\nu+3)})$.

(c) By using the above things, it is clear that we need to show that for $\nu>\frac{1}{2}$, there exists an $x_{\nu-1}>\sqrt{2(\nu+2)}$ such that Q_ν is strictly decreasing on $(0,x_{\nu-1})$ and strictly increasing on $(x_{\nu-1},\infty)$. For this we note that [12, p.276]

$$y_\nu(x)-x\sim-\frac{1}{2}+\frac{4\nu^2-1}{8x}-\cdots,$$

which holds for large values of x and fixed ν. From this we obtain that for $\nu>\frac{1}{2}$ and large values of x we have $y_\nu(x)-x<\nu-1$, or equivalently $Q_\nu(x)<1$. Since Q_ν is strictly decreasing on $(0,\sqrt{2\nu+4})$ and $Q_\nu(x)\to 1$ as $x\to\infty$, it follows that there must be a point at which $Q'_\nu(x)$ vanishes. In other words, there exists an $x_{\nu-1}>\sqrt{2(\nu+2)}$ such that Q_ν is strictly decreasing on $(0,x_{\nu-1})$ and strictly increasing on $(x_{\nu-1},\infty)$. Moreover, Q'_ν changes sign only once, otherwise we will have two local minima between which there must be a local maximum, which is impossible, since in that stationary point $x^\star$ we would have

$$Q''_\nu(x^\star)=\frac{2\nu-1}{(x^\star)^2}\left[R_\nu(x^\star)-\frac{1}{x^\star}\right]>0,$$

according to the above discussion. ■

Proof of Corollary 2.1. Without loss of generality, it is enough to show that inequality (2.2) holds for $b>a>0$. Observe that the function $x\mapsto\mathcal{I}'_\nu(x)$ is convex on $[a,b]$, since as power series has only positive coefficients. Now taking into account that from Theorem 2.1 the function $x\mapsto\mathcal{I}'_\nu(x)$ is strictly log-concave on $(0,\infty)$, from the Hermite–Hadamard inequality [8, p.14] applied to the function $x\mapsto\mathcal{I}'_\nu(x)$ we have

$$\frac{\mathcal{I}_\nu(b)-\mathcal{I}_\nu(a)}{b-a}=\frac{1}{b-a}\int_a^b\mathcal{I}'_\nu(x)\,\mathrm{d}x\geq\mathcal{I}'_\nu\left(\frac{a+b}{2}\right)\geq\sqrt{\mathcal{I}'_\nu(a)\mathcal{I}'_\nu(b)},$$

which completes the proof.

Another proof of Corollary 2.1 is as follows. Let $f:[a,b]\to\mathbb{R}$ be such that $f':[a,b]\to(0,\infty)$ is log-concave. Then, the inequality [13, p.242]

$$\log\left(\frac{1}{b-a}\int_a^b f'(x)\,\mathrm{d}x\right)\geq\frac{1}{b-a}\int_a^b\log f'(x)\,\mathrm{d}x\geq\frac{\log f'(a)+\log f'(b)}{2}\tag{3.3}$$

implies that

$$\log\frac{f(b)-f(a)}{b-a}\geq\log\sqrt{f'(a)f'(b)}$$

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is valid. Applying (3.3) for the function $f = \mathcal{I}_\nu$, which has the property that $x \mapsto \mathcal{I}'_\nu(x)$ is strictly log-concave on $(0, \infty)$, we arrive at (2.2). ■

Proof of Theorem 2.3. By using the recurrence relation [1, p.45] $[x^{-\nu} J_\nu(x)]' = -x^{-\nu} J_{\nu+1}(x)$ we obtain

$$\mathcal{J}'_\nu(x) = -2^\nu \Gamma(\nu+1) x^{-\nu} J_{\nu+1}(x),$$
$$\mathcal{J}''_\nu(x) = 2^\nu \Gamma(\nu+1) x^{-\nu} \left[J_{\nu+2}(x) - \frac{1}{x} J_{\nu+1}(x) \right]$$

and thus the function $x \mapsto -\mathcal{J}'_\nu(x)$, which maps $(0, j_{\nu+1,1})$ into $(0, \infty)$, is strictly log-concave if and only if the function $x \mapsto 1/x - J_{\nu+2}(x)/J_{\nu+1}(x)$ is strictly decreasing on $(0, j_{\nu+1,1})$. But, using the well-known Mittag-Leffler expansion

$$\frac{J_{\nu+1}(x)}{J_\nu(x)} = \sum_{n \geq 1} \frac{2x}{j_{\nu,n}^2 - x^2},$$

we obtain that

$$\left[\frac{J_{\nu+1}(x)}{J_\nu(x)} \right]' = 2 \sum_{n \geq 1} \frac{j_{\nu,n}^2 + x^2}{(j_{\nu,n}^2 - x^2)^2} > 0$$

for all $\nu > -1$, $x \neq j_{\nu,n}$, $n \in \{1, 2, \ldots\}$. Note that this can be proved also by using Neumann's formula for the product of two Bessel functions of the first kind with different orders, see [14, Lemma 2.5]. In particular, the above result means that the function $x \mapsto J_{\nu+1}(x)/J_\nu(x)$ is strictly increasing on $(0, j_{\nu,1})$ for $\nu > -1$. Thus, for $\nu > -2$, the function $x \mapsto 1/x - J_{\nu+2}(x)/J_{\nu+1}(x)$ is strictly decreasing on $(0, j_{\nu+1,1})$ as the sum of two strictly decreasing functions. ■

Proof of Corollary 2.4. Observe that as in Corollary 2.1, it is enough to consider the case when $b > a$. Then, the proof of the result follows from Theorem 2.2 and (3.3) by choosing for f the function $-\mathcal{J}_\nu$. ■

Proof of Theorem 2.5. By using the recurrence relation [1, p.79] $xK'_\nu(x) + \nu K_\nu(x) = -xK_{\nu-1}(x)$, we obtain that

$$\mathcal{K}'_\nu(x) = -2^{\nu-1} \Gamma(\nu) x^\nu K_{\nu-1}(x),$$
$$\mathcal{K}''_\nu(x) = -2^{\nu-1} \Gamma(\nu) x^\nu \left[\frac{\nu}{x} K_{\nu-1}(x) + K'_{\nu-1}(x) \right]$$

and thus $x \mapsto -\mathcal{K}'_\nu(x)$ is strictly log-concave on $(0, \infty)$ if and only if the function

$$x \mapsto \frac{\mathcal{K}''_\nu(x)}{\mathcal{K}'_\nu(x)} = \frac{\nu}{x} + \frac{K'_{\nu-1}(x)}{K_{\nu-1}(x)} = \frac{1}{x} - \frac{K_{\nu-2}(x)}{K_{\nu-1}(x)}$$

is strictly decreasing on $(0, \infty)$. By using [14, Lemma 2.4] the function $x \mapsto K_{\nu+1}(x)/K_\nu(x)$ is strictly decreasing on $(0, \infty)$ for $\nu > -\frac{1}{2}$, which implies that the function $x \mapsto K_{\nu-2}(x)/K_{\nu-1}(x)$ is strictly increasing on $(0, \infty)$ for $\nu > \frac{3}{2}$. Consequently, the function $x \mapsto -\mathcal{K}'_\nu(x)$ is strictly log-concave on $(0, \infty)$ for $\nu > \frac{3}{2}$. Finally, observe that by using (2.10), we have that $x \mapsto -\mathcal{K}'_{3/2}(x) = \frac{\pi}{2} x e^{-x}$ is also strictly log-concave on $(0, \infty)$. This completes the proof. ■

Proof of Corollary 2.6. Clearly, as in Corollary 2.1, it is enough to consider the case when $b > a$. Then, the proof of the result follows from Theorem 2.3 and (3.3) by choosing for f the function $-\mathcal{K}_\nu$. ■

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4. Concluding remarks

In this section, we would like to comment the main results of this paper.

(1) We note that in [6, Theorem 3.6.3] (see also [7, Theorem 3.25]), it was proved that the function $x \mapsto \mathcal{I}'_\nu(x)$ is strictly log-concave on $(0, \sqrt{2(\nu+2)})$ for $\nu > -1$. Moreover, it was shown in [6, Theorem 3.6.4] (see also [7, Theorem 3.26]) that the inequality (2.2) is valid for all $\nu > -1$ and $a, b \in (0, \sqrt{2(\nu+2)})$. Theorem 2.1 and Corollary 2.1 complement and improve the above-mentioned results.

(2) We also note that (see [6, Theorem 3.6.3] or [7, Theorem 3.25]) for $\nu > -1$ the function $x \mapsto \mathcal{I}_\nu(x)$ is strictly log-convex on $(0, \sqrt{2(\nu+1)(\nu+2)})$. Since the function $x \mapsto \log \mathcal{I}_\nu(x)$ is even it follows that in fact $x \mapsto \mathcal{I}_\nu(x)$ is strictly log-convex on $(-\sqrt{2(\nu+1)(\nu+2)}, \sqrt{2(\nu+1)(\nu+2)})$ for $\nu > -1$. Moreover, it was shown in [15, Theorem 4] that the function $x \mapsto \mathcal{I}_\nu(x)$ is strictly log-convex on $[-j_{\nu,1}, j_{\nu,1}]$ for $\nu > -1$, where $j_{\nu,1}$ is the first positive zero of the Bessel function of the first kind J_ν. Next, we show that the function $x \mapsto \mathcal{I}_\nu(x)$ cannot be strictly log-convex on the whole $\mathbb{R}$ for $\nu > -1$. When $\nu \geq -\frac{1}{2}$ the strict log-convexity of $x \mapsto \mathcal{I}_\nu(x)$ is certainly valid, but in the case when $\nu \in (-1, -\frac{1}{2})$, there exists a $z_\nu \geq j_{\nu,1}$ such that the function $x \mapsto \mathcal{I}_\nu(x)$ is strictly log-convex on $(0, z_\nu)$ and strictly log-concave on (x_ν, ∞). For this observe that the function $x \mapsto \mathcal{I}_\nu(x)$ is strictly log-convex (log-concave) if and only if $x \mapsto R_\nu(x)$ is strictly increasing (decreasing). But, it was shown in [9, p.446] that when $\nu \in (-1, -\frac{1}{2})$, the function $x \mapsto R_\nu(x)$ on $(0, \infty)$ is increasing first to reach a maximum and then decreasing. This means that there exists a z_ν such that $x \mapsto R_\nu(x)$ is strictly increasing on $(0, z_\nu)$ and strictly decreasing on (z_ν, ∞), as we required. Now, since $x \mapsto \mathcal{I}_\nu(x)$ is strictly log-convex on $(0, j_{\nu,1}]$ we clearly have that $z_\nu \geq j_{\nu,1}$.

(3) We would like to mention that the idea of the first proof of (2.2) does not work in the case of (2.6) because in view of $\mathcal{J}_{-1/2}(x) = \cos x$, the function $x \mapsto -\mathcal{J}'_{-1/2}(x) = \sin x$ is concave and log-concave and consequently

$$\mathcal{J}'_{-1/2}\left(\frac{a+b}{2}\right) \leq \frac{1}{b-a}\int_a^b \mathcal{J}'_{-1/2}(x)\,\mathrm{d}x = \frac{\mathcal{J}_{-1/2}(b) - \mathcal{J}_{-1/2}(a)}{b-a},$$
$$-\mathcal{J}'_{-1/2}\left(\frac{a+b}{2}\right) \geq \sqrt{\mathcal{J}'_{-1/2}(a)\mathcal{J}'_{-1/2}(b)}$$

and these together does not imply van der Corput's inequality (2.5). However, as we have seen in the proof of Corollary 2.2, the inequality (3.3) applied to the function $-\mathcal{J}_\nu$ implies the van der Corput inequality (2.6). In other words, there is no need to assume for f' to be convex, its log-concavity is quite enough, according to (3.3).

(4) Finally, we would like to mention that functions which have log-concave derivatives occur, for example, in probability theory. Namely many probability density functions are log-concave (see, e.g. [16]), and thus applying the inequality (3.3) for the cumulative distribution functions we can obtain van der Corput inequalities. For example, consider the probability density and cumulative distribution function of the standard normal distribution

$$\varphi(x) = \frac{1}{\sqrt{2\pi}}\mathrm{e}^{-x^2/2} \quad \text{and} \Phi(x) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{x} \mathrm{e}^{-t^2/2}\,\mathrm{d}t.$$

The function $x \mapsto \Phi'(x) = \varphi(x)$ is log-concave on $\mathbb{R}$, and applying the inequality (3.3) for the function $f = \Phi$ we obtain the following van der Corput inequality for all real a and b

$$|\Phi(a) - \Phi(b)| \geq |a-b|\sqrt{\varphi(a)\varphi(b)}.$$

Another example is the gamma distribution which has support $(0,\infty)$, probability density and cumulative distribution function as follows:

$$f(x)=\frac{x^{\alpha-1}\mathrm{e}^{-x}}{\Gamma(\alpha)} \quad \text{and} \quad F(x)=\frac{\gamma(\alpha,x)}{\Gamma(\alpha)}=\frac{1}{\Gamma(\alpha)}\int_0^x t^{\alpha-1}\mathrm{e}^{-t}\,\mathrm{d}t,$$

where $\gamma(\alpha,\cdot)$ is the lower incomplete gamma function and $\alpha>0$ is the shape parameter. Since the probability density function of the gamma distribution is log-concave on $(0,\infty)$ for all $\alpha\geq 1$ (see, e.g. [16, p.192]), applying (3.3) for the function F we obtain the following van der Corput inequality

$$|\gamma(\alpha,a)-\gamma(\alpha,b)|\geq |a-b|(ab)^{(\alpha-1)/2}\mathrm{e}^{-(a+b)/2},$$

where $a,b\geq 0$ and $\alpha\geq 1$.

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